

$$\begin{array}{ccc}
 \mathbf{F}(A) & \xrightarrow{\eta_A} & \mathbf{G}(A) \\
 \mathbf{F}(\phi) \downarrow & & \downarrow \mathbf{G}(\phi) \\
 \mathbf{F}(B) & \xrightarrow{\eta_B} & \mathbf{G}(B)
 \end{array}$$